Math 105L **Lecture 5-2**

The Derivative Function & Its Interpretation

1. The Derivative as a Function

Calculating the derivative point-by-point (e.g. at $a = 1, 2, 4$) using limits involves repeating the exact same algebra each time. Instead, we calculate the derivative for an arbitrary variable x, creating the **derivative function** $f'(x)$:

Definition of the Derivative Function:
Given a function $f(x)$, its derivative function is defined by the limit:

$$\mathbf{f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}}$$

Equivalent Notations for the Derivative of $y = f(x)$:

$$\mathbf{f'(x) = y' = \frac{dy}{dx} = \frac{d}{dx}[f(x)]}$$

2. Qualitative Rules for Graphing $f'(x)$ from $f(x)$

Since $f'(a)$ represents the slope of the tangent line to $y = f(x)$ at $x = a$:

- $f(x)$ **is increasing** $\implies$ slopes are positive $\implies$ $\mathbf{f'(x) > 0}$ (graph of f' lies **above** the x-axis).
- $f(x)$ **is decreasing** $\implies$ slopes are negative $\implies$ $\mathbf{f'(x) < 0}$ (graph of f' lies **below** the x-axis).
- **Local Extrema (Peaks and Valleys):** At turning points where $f(x)$ transitions from increasing to decreasing (or vice versa), the tangent line is horizontal $\implies$ slope is 0 $\implies$ $\mathbf{f'(a) = 0}$ (x**-intercepts** of f').
- **Steepness:** Where $f(x)$ is steepest, $f'(x)$ reaches its maximum or minimum values. Where $f(x)$ flattens out, $f'(x)$ approaches 0.

3. Applied Interpretation & Units of the Derivative

General Rule for Units:
The units of $f'(x)$ are always:

$$\text{Units of } \mathbf{f'(x)} = \frac{\text{Units of output } (\mathbf{y})}{\text{Units of input } (\mathbf{x})}$$

Worked Classroom Example: Eastbound Vehicle Motion

- **Context:** Let $s(t)$ represent the distance due East from Durham of a car traveling along a straight highway, where t is measured in hours and s is measured in miles.
- **Units of $s'(t)$:**

$$\text{Units} = \frac{\text{miles}}{\text{hour}} \quad \text{(instantaneous velocity)}$$

- **Physical Meaning of the Sign of $s'(t)$:**
  - $\mathbf{s'(t) > 0}$: Distance East is increasing $\implies$ The car is traveling **East** (away from Durham).
  - $\mathbf{s'(t) < 0}$: Distance East is decreasing $\implies$ The car is traveling **West** (back toward Durham or westbound).
  - $\mathbf{s'(t) = 0}$: The car is **instantaneously stationary** (changing direction or at rest).
- **Synthesis:** Knowing the formula for $s'(t)$ gives the car's instantaneous speed and directional velocity at any exact hour t.